

Configuration Parameter	Parameter Scope	Exact Value / Implementation Setting
Target Dataset & Split	Global / Fixed	AAMI EC57 Inter-Patient DS1 Partition
CV Strategy	Global / Fixed	3-Fold Patient-Level Cross-Validation
Fold 2 Validation Patients	Fold 2 Specific	['208', '209', '114', '115'] (4 Patients)
Fold 2 Training Patients	Fold 2 Specific	18 remaining DS1 Patients
Sampling Alpha (α)	Global / Fixed	0.7320
Learning Rate (LR)	Global / Fixed	0.001184
Weight Decay (WD)	Global / Fixed	7.114476e-04
Batch Size	Global / Fixed	128
Epochs	Global / Fixed	20 per fold
Loss Function	Global / Fixed	CrossEntropyLoss(ignore_index=4)
Optimizer	Global / Fixed	Adam
Model Architecture	Global / Fixed	GenericHybrid1DBiCNNGRU
Trainable Parameters	Global / Fixed	578,309 (Control) / 257,541 (Medium)
WeightedRandomSampler	Global / Fixed	Enabled for Training (replacement=True)
Sampler Weight Formula	Global / Fixed	$w_c = (1 / N_c)^{0.7320}$ for $c \in \{0..3\}$, $w_Q = 0$
Validation Sampler	Global / Fixed	Unweighted (Natural Class Distribution)
Fold 2 Active Train Samples	Fold 2 Specific	40,021 (N: 36,842 SVEB: 546 VEB: 2,111 F: 522)
Fold 2 Active Val Samples	Fold 2 Specific	9,779 (N: 7,971 SVEB: 397 VEB: 1,035 F: 376)
Fold 2 Class Weights (w_c)	Fold 2 Specific	w_N : 4.54e-4, w_{SVEB} : 9.90e-3, w_{VEB} : 3.67e-3, w_F : 1.02e-2
Fold 2 Theoretical Props	Fold 2 Specific	N: 41.87%, SVEB: 13.51%, VEB: 19.38%, F: 13.34%, Q: 0.0%